

Assignment 5: Agent Evaluation Dataset

Personal Productivity and Task Execution Agent

Evaluation Metrics

Metric	Formula	Target	Actual	Result
Tool Selection Accuracy	Correct tool selected ÷ tool-applicable cases × 100	≥85%	100%	PASS
Argument Accuracy	Correct arguments ÷ total tool calls × 100	≥80%	100%	PASS
Task Completion Rate	Successful requests ÷ total requests × 100	≥80%	100%	PASS
Approval Compliance	Correct approvals ÷ sensitive actions × 100	100%	100%	PASS
Invalid Action Rate	Invalid actions ÷ total runs × 100	<10%	0%	PASS
Average Response Time	Mean execution time	Record only	Approximately 2–5 seconds	PASS
Recovery Rate	Successfully handled failures ÷ total failures × 100	Record only	100%	PASS

Category A — Direct Response Cases

Requests that should NOT trigger any tool.

#	User Request	Expected Tool(s)	Expected Arguments	Approval	Expected Outcome	Actual Outcome	Pass/Fail	Notes
1	Explain the difference between High and Critical priority tasks.	None	N/A	No	Agent explains directly without using tools.	Agent explained the difference correctly without calling any tool.	PASS	No unnecessary tool call.
2	What does the Blocked status mean?	None	N/A	No	Explain Blocked status.	Correct explanation returned.	PASS	Direct response only.
3	What information do you need to create a task?	None	N/A	No	List required task fields.	Agent listed title, description, priority, due date, tags.	PASS	No tool used.
4	How does the work-plan tool decide what to schedule first?	None	N/A	No	Explain planning logic.	Agent explained priority, due date and workload logic.	PASS	No tool invoked.
5	What is the difference between a task and a reminder?	None	N/A	No	Conceptual explanation.	Correct explanation returned.	PASS	No tool required.
6	What happens if I reject an approval request?	None	N/A	No	Explain rejection behaviour.	Agent explained that the action is cancelled and control returns to the user.	PASS	Correct behaviour.

Category B — Single Tool Cases

#	User Request	Expected Tool(s)	Expected Arguments	Approval	Expected Outcome	Actual Outcome	Pass/Fail	Notes
7	Create a task: Buy groceries, high priority, due tomorrow.	create_task	Title, Priority, Due Date	No	Task created successfully.	Task created successfully with Task ID.	PASS	Verified during testing.
8	Show my high-priority tasks.	list_tasks	Priority=High	No	Display matching tasks.	High-priority task list displayed correctly.	PASS	Correct filtering.
9	Search my notes for marketing campaign.	search_notes	Query="marketing campaign"	No	Matching notes displayed.	Matching note found and displayed correctly.	PASS	Keyword search successful.
10	Create a note: Marketing campaign kickoff meeting is on Monday at 10 AM.	save_note	Title, Content	No	Note saved.	Note saved successfully with Note ID.	PASS	Verified by searching later.
11	Prepare my weekly productivity report.	generate_weekly_report	Current week	No	Weekly report generated.	Report generated showing completed, pending and recommended tasks.	PASS	Correct output.
12	Prepare my work plan for today using a 4-hour workday.	generate_work_plan	Available hours=4	No	Daily work plan generated.	Ordered schedule generated with deferred tasks.	PASS	Priority considered correctly.
13	Create a reminder for Project review on August 5, 2026 at 2 PM.	create_reminder	Title, DateTime	Yes	Approval then reminder created.	Approval requested then reminder created successfully.	PASS	Approval workflow verified.
14	Mark finish report as complete.	complete_task	Task ID	Yes	Approval then task completed.	Approval requested then task marked Completed.	PASS	Completion verified in task list.
15	Update task "Submit assignment" due date to July 15, 2026.	update_task	Task ID, Due Date	Yes	Approval then due date updated.	Task updated successfully.	PASS	Later used for overdue testing.
16	Find overdue tasks and recommend what I should work on first.	list_tasks + priority evaluation	Status/Due Date	No	Identify overdue tasks and recommend priority.	Agent correctly recommended "Submit assignment" because it was overdue and High priority.	PASS	Recommendation matched expected behaviour.

Category C — Multi-Tool Cases

#	User Request	Expected Tool(s)	Expected Arguments	Approval	Expected Outcome	Actual Outcome	Pass/Fail	Notes
17	Create three tasks from these meeting notes.	extract_meeting_actions + create_task	Meeting notes	Yes	Tasks proposed then created after approval.	Tasks extracted, approval requested, tasks created successfully.	PASS	Workflow A verified.

#	User Request	Expected Tool(s)	Expected Arguments	Approval	Expected Outcome	Actual Outcome	Pass/Fail	Notes
18	Draft a follow-up email based on these meeting notes.	draft_follow_up_email	Meeting notes	Yes	Approval then email draft.	Professional email draft generated.	PASS	Approval workflow verified.
19	Create three tasks from these meeting notes: Finish UI, Email client, Prepare slides.	extract_meeting_actions + create_task	Meeting notes	Yes	Extract action items, request approval, create tasks.	Agent extracted three tasks, requested approval, then created all tasks successfully.	PASS	Workflow A completed successfully.
20	Prepare my daily work plan using my pending tasks.	list_tasks + generate_work_plan	Pending tasks, available hours	No	Retrieve pending tasks and create schedule.	Daily work plan generated with ordered schedule and deferred tasks.	PASS	Workflow B verified.
21	Prepare my weekly productivity report.	list_tasks + generate_weekly_report	Current week tasks	No	Generate report with completed and pending tasks.	Weekly report generated correctly.	PASS	Workflow C verified.
22	Find overdue tasks and recommend what I should work on first.	list_tasks + priority_analysis	Task list	No	Detect overdue tasks and recommend priority.	Overdue task detected and recommendation returned.	PASS	Priority logic correct.
23	Search my notes for marketing campaign and summarize the result.	search_notes + summarize_notes	Keyword	No	Retrieve and summarize note.	Matching note found and summarized.	PASS	Multi-tool execution successful.
24	Show my high-priority tasks and mark the second one complete.	list_tasks + complete_task	Priority filter + Task ID	Yes	Retrieve tasks, resolve reference, request approval, complete task.	Agent correctly identified the second task, requested approval, and completed it.	PASS	Session memory verified.
25	Create a reminder for Project review and then draft a follow-up email.	create_reminder + draft_follow_up_email	Reminder details + meeting notes	Yes	Approval before both write operations.	Reminder and email draft created successfully.	PASS	Multiple write operations handled correctly.
26	Convert these meeting notes into tasks and prepare today's work plan.	extract_meeting_actions + create_task + generate_work_plan	Meeting notes	Yes	Create tasks then generate work plan.	Tasks created after approval and work plan generated.	PASS	Complex workflow successful.

Category D — Approval Cases

#	User Request	Expected Tool(s)	Expected Arguments	Approval	Expected Outcome	Actual Outcome	Pass/Fail	Notes
27	Create three tasks in one request.	create_task	Multiple task objects	Yes	Approval card displayed.	Approval requested before execution.	PASS	Requirement satisfied.
28	Mark finish report as complete.	complete_task	Task ID	Yes	Approval before completion.	Approval displayed and task completed after approval.	PASS	Verified.

#	User Request	Expected Tool(s)	Expected Arguments	Approval	Expected Outcome	Actual Outcome	Pass/Fail	Notes
29	Update task “Submit assignment” due date to July 15, 2026.	update_task	Task ID, Due Date	Yes	Approval before update.	Approval requested and task updated.	PASS	Verified.
30	Create a reminder for Project review.	create_reminder	Reminder details	Yes	Approval before reminder creation.	Approval displayed and reminder created.	PASS	Verified.
31	Draft a follow-up email based on meeting notes.	draft_follow_up_email	Meeting notes	Yes	Approval before email generation.	Approval requested and email draft generated.	PASS	Verified.

Category E — Failure & Edge Cases

#	User Request	Expected Tool(s)	Expected Arguments	Approval	Expected Outcome	Actual Outcome	Pass/Fail	Notes
32	(Empty message)	None	N/A	No	Display validation message requesting input.	Agent displayed “I didn’t receive any input — please type a request.”	PASS	Empty input handled correctly.
33	Mark task TASK-123456 as complete.	complete_task	Invalid Task ID	Yes	Display task not found message.	Agent reported that the task ID does not exist.	PASS	Unknown task handled safely.
34	Search my notes for a topic that does not exist.	search_notes	Query	No	No matching notes found.	Agent informed the user that no matching notes were found.	PASS	Graceful failure handling.
35	Use an exhausted/invalid Gemini API key.	LLM API	API Key	No	Show user-friendly API error.	Application displayed an understandable API/quota error without exposing secrets.	PASS	API error handled correctly.

Final Evaluation Metrics

Metric	Formula	Target	Actual	Result
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Invalid Action Rate	Invalid actions ÷ total runs × 100	<10%	0%	PASS
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Conclusion

A total of 35 evaluation test cases were executed across Direct Response, Single Tool, Multi-Tool, Approval, and Failure/Edge scenarios. The Personal Productivity and Task Execution Agent successfully selected the correct tools, validated user inputs, requested approval for sensitive actions, executed workflows accurately, maintained session memory, logged execution history, and handled invalid or unsupported requests gracefully. Based on the completed evaluation, the agent met or exceeded all required performance targets and successfully satisfied the functional requirements of the assignment.